MINI ASSIGNMENT

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Task: Given a support ticket, extract entities - determine urgency - produce final JSON. The assessment specifically contains few-shot in Step 1, CoT in Step 2, and JSON mode in Step 3.

Step – 1 -- Few-shot Entity Extraction

You are a support-ticket entity extractor.

Extract the key entities from the support ticket.

Examples:

Ticket: "My payment failed twice and my account is now locked."

Entities: payment failure, account locked

Ticket: "The mobile app crashes whenever I try to upload a photo."

Entities: mobile app, app crash, photo upload

Now extract the key entities from this ticket:

"{SUPPORT_TICKET}"

Return only a comma-separated list of entities.

Step – 2 -- CoT Urgency Reasoning

You are a senior customer-support triage specialist.

Determine the urgency level of the support ticket using the extracted entities from Step 1.

Urgency levels:

- low: minor issue, inconvenience, or general request

- medium: significant problem affecting the user's normal usage

- high: critical issue involving security, account access, financial loss, or a service outage

Extracted entities:"{STEP_1_OUTPUT}"

Reason step by step about the severity and then give the final urgency level.

Return:

Urgency: low/medium/high

Step – 3 -- JSON Formatting

Return ONLY valid JSON in exactly this structure:

```
{
  "entities": ["entity1", "entity2"],
  "urgency": "low",
  "summary": "short summary of the support issue"
}
```

Entities from Step 1:

{STEP_1_OUTPUT}

Urgency from Step 2:

{STEP_2_OUTPUT}

Original support ticket:

{SUPPORT_TICKET}

Do not add Markdown, explanations, or any text outside the JSON object.

End – to – End Test

Input support ticket: I cannot log into my account. I have tried resetting my password three times, but I never receive the reset email. I need access urgently because I have an important payment due today.

Step 1 Output:

account login, password reset, reset email, account access, payment due

Step 2 Output:

Urgency: high

Step 3 Final Output:


```
{
  "entities": [
    "account login",
    "password reset",
    "reset email",
    "account access",
    "payment due"
  ],
  "urgency": "high",
  "summary": "The user cannot access their account because password reset emails are not being received, with an important payment due today."
}
```